

Formalizing SHIM6, a Proposed Internet Standard in UPPAAL

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FORMALIZING SHIM6

AN IETF PROPOSED INTERNET STANDARD IN UPPAAL

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Background

Multihoming

- A technique to increase the reliability of a network connection.
- Features redundancy, load sharing, performance and policy.
- Current multihoming practices (IPv4) impose a threat on address and routing scalability.
- SHIM6 is a proposal by the IETF to provide multihoming that solve these issues.
- No formal methods have been applied to the draft specification.

Aim; improve the quality of the specification by applying formal methods.

How SHIM6 works

IP roles SHIM6 splits the two semantics of an IP address (*end point identifier* and *locator role*).

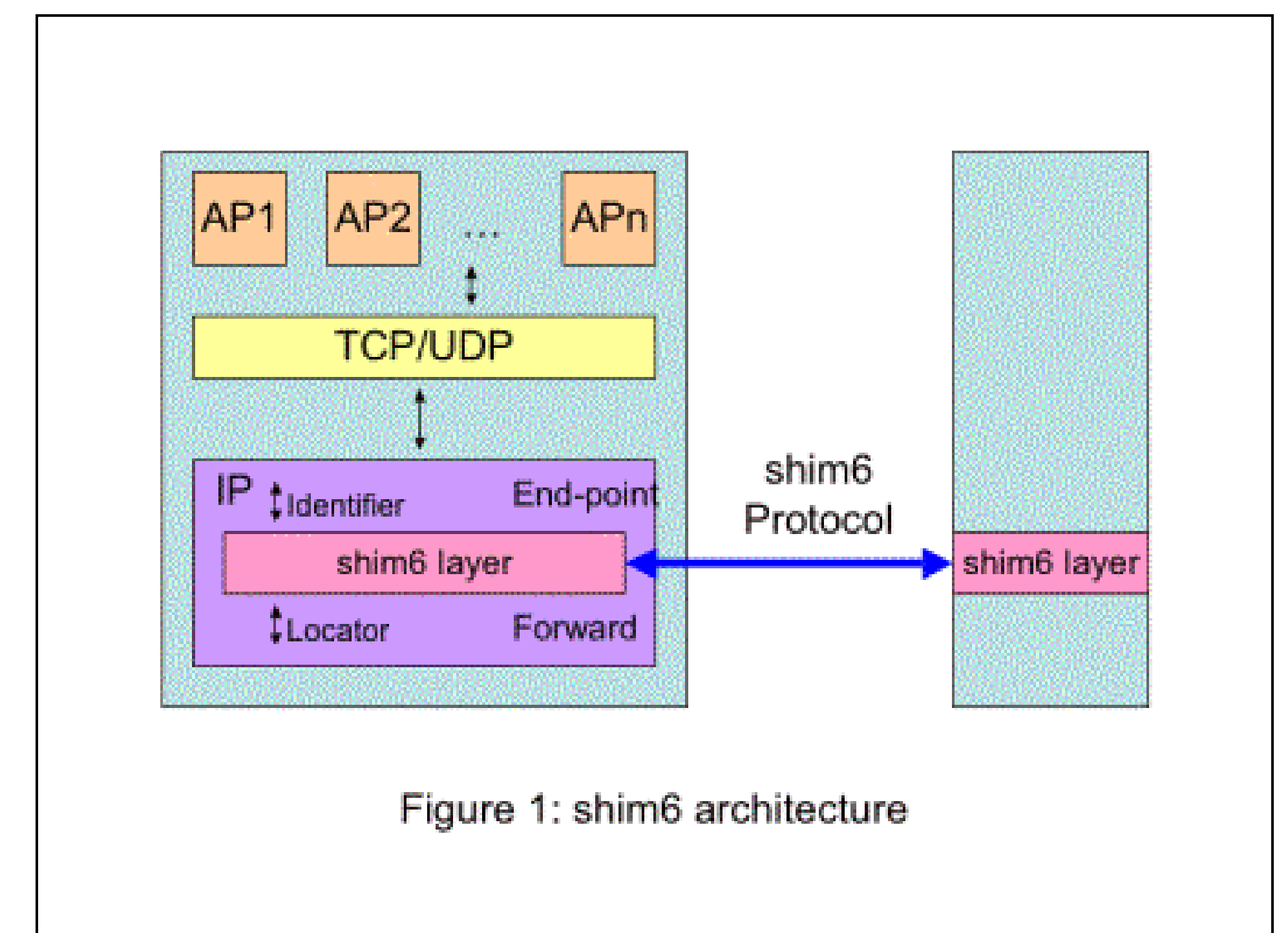
Initial contact Normal data communication between end point identifiers, no SHIM6 needed.

Context Establishment Communication to exchange multihoming information. Data communication remains normal.

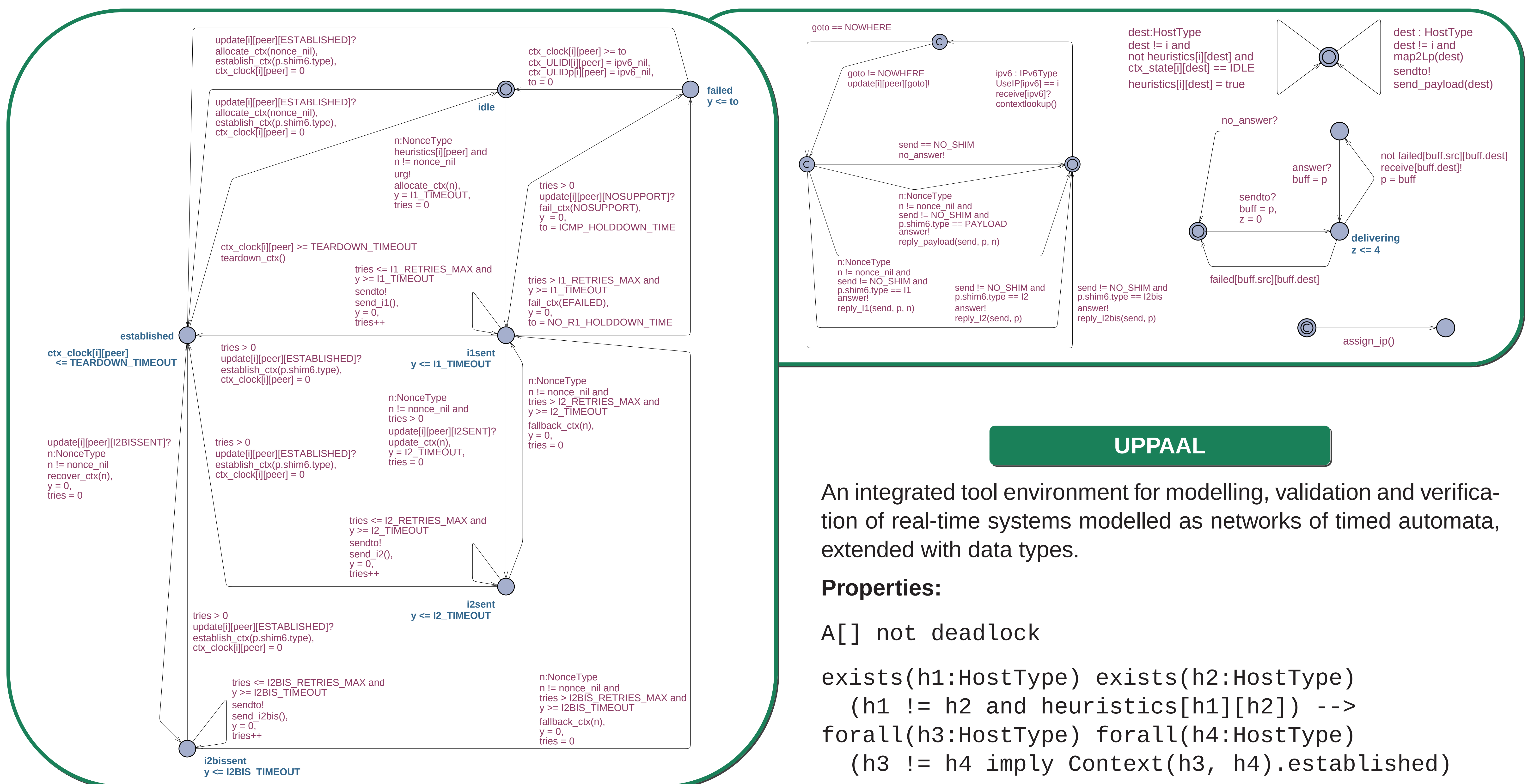
Failure detection Messages are transmitted to detect a link failure.

Locator pair exploration In case of a link failure, a new locator needs to be selected. Locators are mapped back at the host to the

end point identifier. Transport session remains stable. Communication resumes with SHIM6 data packets that provide mapping information.



Formalization and Verification



Results

- Revealed incorrectness upon receiving payload in I2-SENT or I2BIS-SENT.
- Revealed possible deadlock with optional re-transmitting I2 / I2bis messages.
- Clarified confusion about responder nonce.

Revealed several other ambiguities, omissions and inconsistencies. Acknowledged by SHIM6 draft authors. Will be incorporated in new IETF proposal.

Future Work

UPPAAL:

Improve model to verify on scale. Add failure detection and exploration. Extend UPPAAL verifier language. Indicate model state space.

SHIM6:

Implementations. Add HBA and CGA, Context Forking. INTEROP test. Traffic engineering issues.

Further Information

SHIM6:

<http://www.shim6.org>
<http://tools.ietf.org/wg/shim6/>
<http://www.ietf.org/html.charters/shim6-charter.html>

UPPAAL:

<http://www.uppaal.com>

Master Thesis (Approx. May 2007):

<http://www.ita.cs.ru.nl/publications/papers/fvaan/SHIM6/>